Ziyi Han

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Google Scholar: https://scholar.google.com/citations?user=u-lzxf8AAAAJ&hl=zh-CN&oi=ao

https://ziyihanzyh.github.io/

Research Interests

Online Learning, Large Language Models, Optimization Algorithm Design — My previous research focused on edge intelligence, particularly on optimizing model training and inference and GPU resource sharing through online learning methods. Currently, my academic interests center on online learning and large language models (LLMs). I am especially interested in how online learning can enable LLMs to adapt continuously from interactive data, and conversely, how LLMs can support more intelligent and data-efficient online learning frameworks.

Education

The Chinese University of Hong Kong — 08/2024 – present
Doctor of Philosophy — Hong Kong, China
- Department of Computer Science and Engineering
- Supervisor: John C.S. Lui

Wuhan University — 09/2021 – 06/2024
Master of Engineering — Wuhan, China
- School of Cyber Science and Engineering
- Supervisor: Ruiting Zhou
- GPA: 3.57/4.0

Wuhan University — 09/2017 – 06/2021
Bachelor of Engineering — Wuhan, China
- School of Cyber Science and Engineering
- GPA: 3.81/4.0
- rank: 9/106

Publications

Published Papers

* is the corresponding author

- **Ziyi Han**, Ruiting Zhou*, Chengzhong Xu, Yifan Zeng, and Renli Zhang InSS: An Intelligent Scheduling Orchestrator for Multi-GPU Inference with Spatio-Temporal Sharing, IEEE Transactions on Parallel and Distributed Systems (**TPDS**), 2024, to appear. (**CCF A**)
- Ruiting Zhou, **Ziyi Han***, Yifan Zeng, Zhi Zhou, Libing Wu, Wei Wang. SAFE: Intelligent Online Scheduling for Collaborative DNN Inference in Vehicular Network, in Proc. of **CSCWD** 2024, Tianjin, China, May 8-10, 2024. (**CCF C**)
- **Ziyi Han**, Ruiting Zhou*, Jinlong Pang, Haisheng Tan*, Yue Cao. Online Scheduling Unbiased Distributed Learning over Wireless Edge Networks, in Proc.of IEEE **ICPADS**, Beijing, China, Dec.13-15, 2021. (**CCF C**, **Outstanding Paper Award**)
- Yifan Zeng, Ruiting Zhou*, Lei Jiao, **Ziyi Han**, Jieling Yu. Efficient Online DNN Inference with Continuous Learning in Edge Computing, in Proc. of IEEE/ACM **IWQoS** 2024, Guangzhou, China, Jun. 19-21, 2024. (**CCF B**)

- Jinlong Pan, **Ziyi Han**, Ruiting Zhou*, Renli Zhang, John C.S. Lui and Hao Chen. Eris: An Online Auction for Scheduling Unbiased Distributed Learning over Edge Networks. IEEE Transactions on Mobile Computing (**TMC**), 23(6):7196-7209, June 2024. (**CCF A**)
- Jinlong Pang, **Ziyi Han**, Ruting Zhou*, Haisheng Tan, Yue Cao. Online Scheduling Algorithms for Unbiased Distributed Learning over Wireless Edge Networks. Journal of Systems Architecture (**JSA**), 131,1-14, Oct. 2022. (**CCF B**)

Papers in Submission/Revision

- **Ziyi Han**, Huanyu Wang, Zeyu Zhang, Xiangxiang Dai, Xutong Liu, John C.S. Lui. HiLoRA: Adaptive Hierarchical LoRA Routing for Training-Free Domain Generalization. arXiv preprint arXiv:2510.12266, 2025. (submitted to ICLR 2026)
- **Ziyi Han**, Xutong Liu, Ruiting Zhou, Xiangxiang Dai, John C.S. Lui. Faster, Smaller, and Smarter: Task-Aware Expert Merging for Online MoE Inference. arXiv preprint arXiv:2509.19781, 2025. (submitted to INFOCOM 2026)
- **Ziyi Han**, Ruiting Zhou, Haisheng Tan, John C.S. Lui. Online Scheduling with Trajectory Prediction for Collaborative DNN Inference in Vehicular Networks. IEEE Transactions on Networking (**ToN**). (Major Revision)

Research Experiences

Research Intern at Huawei 2012 Lab, Hangzhou, China

Leader: Prof. Haibo Chen 06/2025-08/2025

- Conducted research on efficient domain generalization for large language models (LLMs), focusing on reusing open-source LoRA adapters across heterogeneous tasks without retraining.
- Proposed a training-free hierarchical routing framework that dynamically activates task-relevant LoRAs and rank-one components (ROCs) to enhance adaptability and efficiency.
- Developed theoretical error bounds guaranteeing robust routing for both seen and unseen domains. Implemented large-scale experiments on LLaMA2-7B and FLAN-T5-large, showing great improvement in accuracy with moderate inference cost.

Honors and Scholarships

- First Prize of Cybersecurity Excellence Scholarship 2021-2023
- First Prize Scholarship of Academic Excellence 2022
- Huawei Scholarship 2022
- Wuhan University 2021 Outstanding Bachelor's Degree Thesis 2021
- Third Prize of University Scholarship 2020

Skills

Programming Languages

Python, C++, MATLAB

Language Proficiency

English (IELTS: 7.0, CET-6: 552), Chinese (native)